

TABULATED DELAYED NEUTRON GROUP PARAMETERS FOR MOLTEN SALT REACTORS

BY

LUKE SEIFERT

DISSERTATION

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Professor Kathryn D. Huff, Advisor
More

Abstract

Beta-delayed neutron emitters, or delayed neutron precursors, are important for safeguards monitoring and safety considerations in reactor transients, including shutdown decay-heat, reactor controllability, and reactivity changes. In particular, fluid-fueled molten salt reactors (MSRs) have delayed neutron precursors (DNPs) which move from areas of high fission in the core to areas of lower fission in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. Many works have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. However, simulations that couple depletion of the fuel salt while accounting for fuel movement are computationally-intractable [13, 95]. Because of this limitation, DNP group parameters used in MSR dynamics simulations use fresh fuel salt and do not account for depletion. This issue is important to resolve, as the DNP group parameters are used for safety-related calculations as well as in safeguards-related detection simulations.

The DNP group parameters for MSRs can be resolved by first generating concentrations that account for the spatially resolved depletion and the continuous online reprocessing. Shorter lived DNP group parameters are found via pulse irradiation experiments, which occur at very short time scales relative to molten salt flow rates. The longer lived DNP group parameters are expected to vary more since they are measured with saturation, or infinite, irradiation experiments which occur at time scales several times longer than the longest DNP half life. These longer time scales mean that thermal-hydraulic, thermochemical, and reprocessing effects are able to affect the DNPs.

The concentrations are combined with delayed neutron emission probability and half life data for each nuclide to construct delayed neutron counts as a function of time. A non-linear least-squares solver is then used in order to generate a χ^2 minimized set of DNP group parameters. These updated MSR DNP group parameters are then used in several safety-related transient and safeguards simulations in order to demonstrate the difference between the group parameters. This work solves an important problem with the DNP group parameters for use in MSRs by providing updated group parameters and a method for generating them dynamically for any given reactor. Tabulated DNP group parameters for various flow rates and energies are provided in this work for future analyses.

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Chapter 1

Introduction

Overview of rest of document

In XXXX, Keepin used a sample of U235 in Godiva to generate DNP group parameters. In 1988, Brady's thesis used updated data on 271 nuclides identified as DNPs. This data was used for ENDF-B/VI was released, and since then there have not been any changes to the DNP group parameters (cite the different releases of endf).

Brady's thesis has a good history (up to 1988).

This document should contain tabulated data for various fissile nuclides, neutron irradiation energies, in-core residence times, ex-core residence times, and mixing rates. Additionally, data should be provided on the exact concentrations of each nuclide (up to 99% of the DN yield).

Chapter 2

Background and Motivation

This work aims to generate more accurate DNP group parameters and implement those parameters in safeguards and safety-related transient scenarios. Thus, a background is necessary to properly motivate these aspects of this work. Nuclear safeguards and safety-related reactor transients are the primary motivators behind this work, so they are discussed first. Various information relating to MSRs is then provided, including: depletion simulations, DNP treatment, thermochemistry, and the specific reactor analyzed in this work.

2.1 Nuclear Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Within nuclear safeguards there are two general categories: proliferation resistance (PR) and physical protection (PP) [78]. This work focuses on PR, which is defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices." [78] This definition agrees with the definition made at the International Atomic Energy Agency (IAEA) international workshop in 2002 [52].

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [78]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability is 0-5 percent, while a very high PR detection probability is 95-100 percent [78].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) Order in February of last year that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [48]. Based on the more recent Order, SNM is classified as shown in Table 2.1.

Table 2.1: SNM classification table recreated from [48].

Material Type	Accountable Quantity	Weight Field Used for Element	Weight Field Used for Isotope
U-235	1 g	total U	U-235
U-233	1 g	total U	U-233
Pu-242	1 g	total Pu	Pu-242
Pu-239-241	1 g	total Pu	Pu-239 + Pu-241
Pu-238	0.1 g	total Pu	Pu-238
Am-241**	1 g	total Am	Am-241
Am-243**	1 g	total Am	Am-243
Np-237**	1 g	total Am	Np-237

** Americium and neptunium are treated as SNM if they are separated.

Table 2.1 is used along with the graded safeguards table in order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A" to "E". High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% uranium-235 or <10% uranium-233.

The quantities provided in Table 2.1 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. If the material is classified as attractiveness level "C", then any quantity of plutonium or uranium-233 ≥ 6 kg is considered Category I. If the material contained uranium-235, separated neptunium-237, or americium-241 and 243, then an amount ≥ 20 kg is considered Category I.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" does not have a listed Category I quantity for material containing uranium-235, separated neptunium-237, or americium-241 and 243.

Material of attractiveness level "E" does not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over 20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). Significant quantities of the different material categories are as follows [78, 3, 10]:

- Direct use
 - 8kg of Pu
 - 25kg of HEU
 - 8kg of ^{233}U
- Indirect use
 - 75kg of ^{233}U
 - 10t of natural U
 - 20t of depleted U

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8kg of plutonium as a significant quantity. If that fuel salt also had an attractiveness level of "D", then 16kg of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. Detection of these diversion scenarios is then investigated via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change.

2.2 Safety-Related Reactor Transients

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients are caused by postulated initiating events (PIEs), which are defined by the IAEA in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs which lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there is not specific events listed for MSRs, guidance is provided on what sort of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, events which are not independent of each other could potentially occur at the same time.

Safety-related transients, for MSRs include unprotected loss of heat sink (ULOHS), unprotected loss of flow (ULOF), unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [65, 27]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, which causes a loss of cooling of the reactor. For the Molten Salt Fast Reactor (MSFR), Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [20]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

These safety-related transients are important for the current work, as the tabulated DNP group parameters can be used to more rapidly simulated these transients. Additionally, due to the coupling of the neutronics and thermal-hydraulics in calculation of the tabulated effective group parameters, the results are more accurate than they would be using more traditional transient analysis methods.

2.2.1 Transient Analysis Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include, as of 2018, "...fluid-flow-velocity field coupled multi-group, time-resolved, neutron diffusion; ... point reactor kinetics schemes; and quasi-statics; generally all coupled, tightly or loosely, to a thermal-hydraulics solver" [101].

Multi-group diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches which can be applied to transient problems. Hurwitz and Ehrlich have a general formulation of multi-group diffusion written in 1953, and many more recent works have used the method [50].

Examples of works that use fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [38, 95]. Other examples can be found in the review by Wooten et al. [57, 39, 56]. Even more works which use the multi-group diffusion method can

be found in the literature [105, 103, 58, 74]

These works typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling can be seen in Equations (2.1) and (2.2) [101].

$$\begin{aligned} \frac{1}{\nu_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned} \quad (2.1)$$

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\nu \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i \quad (2.2)$$

The terms in Equations (2.1) and (2.2) are defined as follows:

ν_g is the average number of neutrons produced by fission of group g

ϕ_g is the scalar neutron flux of group g

D_g is the group g diffusion term

G is the total number of energy groups

$\Sigma_{s,g' \rightarrow g}$ is the macroscopic scattering cross section from group g' to group g

β is the delayed neutron fraction

$\chi_{p,g}$ is the number of prompt neutrons born in group g

Σ_f is the macroscopic fission cross section

I is the number of delayed neutron precursor groups

$\chi_{d,g,i}$ is the number of delayed neutrons from precursor group i born in group g

λ_i is the decay constant for group i

C_i is the concentration of group i

$\vec{u}$ is the velocity profile

$\Sigma_{a,g}$ is the macroscopic absorption cross section for group g

β_i is the delayed neutron fraction for group i

ν_T is the eddy viscosity

Sc_T is the turbulent Schmidt number

Modified point kinetics

Point reactor kinetics (PRK) are a simpler approach to modeling a transient, with the basic form given in Equations (2.3) and (2.4). The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [46, 60]. modified point kinetics (MPK) are used for MSRs, as standard PRK do not properly handle the movement of the fuel salt.

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t) \quad (2.3)$$

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t) \quad (2.4)$$

The terms in Equations (2.3) and (2.4) are defined as follows:

n is the neutron density

ρ is the reactivity

β_{eff} is the effective delayed neutron fraction

Λ is the prompt neutron generation time

λ_i is the group i decay constant

C_i is the concentration of group i

$\beta_{eff,i}$ is the effective delayed neutron fraction from group i

PRK in general struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK does, however, give a correct prediction for general evolution trends [101].

Wooten et al. provide four different works which use MPK in MSR analysis [43, 71, 93, 104].

The work by Guo et al. analyzes ULOHS and ULOF transients in Molten Salt Actinide Recycler and Transmuter (MOSART). The PRK equations are modified into the delayed point reactor kinetics (DPK), which model precursor

concentrations in-core and ex-core. These equations, shown in Equations (2.5) through (2.7) (recreated from [43, 93]), are also used by Shi et al. in Reactor Excursion and Leak Analysis Program (RELAP) for a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [93]. This approach enables modeling of ex-core decay of DNPs, capturing effects at power and for loss of flow, but "...may fail to adequately capture reactivity oscillations in transients occurring at low or zero power and with increases in flow rate" [101].

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,core}(t) \quad (2.5)$$

$$\begin{aligned} \frac{dC_{i,core}(t)}{dt} = & \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_{core}} \right) C_{i,core}(t) \\ & + \frac{1}{\tau_{loop}} \left(\frac{V_{loop}}{V_{core}} \right) C_{i,loop}(t - \tau_{loop}) \end{aligned} \quad (2.6)$$

$$\frac{dC_{i,loop}(t)}{dt} = \frac{1}{\tau_{core}} \left(\frac{V_{core}}{V_{loop}} \right) C_{i,core}(t) - \left(\lambda_i + \frac{1}{\tau_{loop}} \right) C_{i,loop}(t) \quad (2.7)$$

Intermediate terms used in Equations (2.5) through (2.7) are shown in Equations (2.8) through (2.11).

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss} \quad (2.8)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,loop} \quad (2.9)$$

$$\tau_{core} = \frac{M_{core}}{w(t)} \quad (2.10)$$

$$\tau_{loop} = \frac{M_{loop}}{w(t)} \quad (2.11)$$

The terms in Equations (2.5) through (2.11) are defined as follows, with most terms identical to those in the PRK equations:

V_{loop} is the volume of the loop

V_{core} is the volume of the core

β_{static} is the value of β_{eff} without flow

M_{core} is the mass in the core

M_{loop} is the mass in the loop

$w(t)$ is the mass flow rate

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [94]. That work uses a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. uses a method called "IPK method," which may stand for "Improve Point-Kinetic method," though this is not explicitly stated [71]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes "a uniform fission distribution in the core, together with an instantaneous propagation and extraction of the heat produced" [71].

The work by Zanetti et al. creates an extension of the TRACE code, a finite volume code used for light water reactor (LWR) transients [104]. The DNP concentration as a function of space and time is resolved upwind in space and with backward Euler for time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to PRK equation, and the set of DNP concentration and the PRK power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [61]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich, and used in PRK equations [45].

Quasi-statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [101]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [46, 30]. There are also other works which discuss the quasi-static method in detail [11].

Wooten et al. introduce several different works which have used quasi-statics [101]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [80].

The neutron flux is treated as shown in Equation (2.12), where there is a flux amplitude, $N(t)$, which is recalculated more frequently than the flux shape, $\Phi(\vec{x}, t)$. The neutron flux is constrained in the work by a term containing the phase-space integration and the weighting function used.

$$\Phi(\vec{x}, t) = N(t)\psi(\vec{x}, t) \quad (2.12)$$

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the "shape" steps are recreated here in Equations (2.13) and (2.14). These equations are solved by time-discretizing Equation (2.13) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5].

$$\begin{aligned} \frac{1}{v} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{vN} \frac{dN}{dt} \psi(\vec{x}, t) + M(t)\psi(\vec{x}, t) = \\ (1 - \beta)\chi_p F(t)\psi(\vec{x}, t) + \frac{1}{N} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{N} Q(\vec{x}, t) \end{aligned} \quad (2.13)$$

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = N\beta_i F\psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(u(\vec{x}, t)C_i(\vec{x}, t)) \quad (2.14)$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters are calculated as well as the flux amplitude. The equations used in the "reactivity" steps are recreated here in Equations (2.15) through (2.18).

$$\frac{dN}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) N + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}) \quad (2.15)$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} N - \lambda_i C_{s,i} \quad (2.16)$$

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle} \quad (2.17)$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d C_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle} \quad (2.18)$$

Intermediate terms used in Equations (2.15) through (2.18) are shown in Equations (2.19) through (2.20). The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.13) and (2.14) to zero as

well as the fluid velocity to zero at $t=0$. This allows Equation (2.19) to be solved.

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla \cdot (u(\vec{x}, t) C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla \cdot (u(\vec{x}, t) C_{m,i}(\vec{x}, t)) \quad (2.19)$$

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = N(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i \partial C_{s,i}(\vec{x}, t) \quad (2.20)$$

The terms in Equations (2.12) through (2.20) are defined as follows:

Φ is the neutron flux

$N(t)$ is the neutron flux amplitude

$\psi(\vec{x}, t)$ is the neutron flux shape

v is the average neutron speed

$M(t)$ is the neutron disappearance operator

$F(t)$ is the neutron production operator

β is the delayed neutron fraction

β_{eff} is the effective delayed neutron fraction

χ_p is the prompt neutron energy spectra

χ_d is the delayed neutron energy spectra

χ is the neutron energy spectra

λ_i is the decay constant for DNP group i

$C_i(\vec{x}, t)$ is the concentration of DNP group i

$Q(\vec{x}, t)$ is the external neutron source

$u(\vec{x}, t)$ is the DNP velocity

$C_{s,i}(\vec{x}, t)$ is the "standard" precursor concentration of group i

$C_{m,i}(\vec{x}, t)$ is the "movable" precursor concentration of group i

ρ is the reactivity

$c_{s,i}$ is the PRK "standard" precursor concentration of group i

$c_{m,i}$ is the PRK "movable" precursor concentration of group i

q_{eff} is the PRK external neutron source

Λ is the prompt neutron generation time

$W(x)$ is the weighting function (by default in SIMMER is an adjoint flux computed at steady-state conditions)

The general formulation used for quasi-static methods is very similar among the various works that use it [34, 95].

Alternative Methods

Although multi-group diffusion, PRK, and quasi-statics are the most commonly used methods, there are other choices used to analyze transients. One of these is the Transient Fission Matrix (TFM) method, which entails generating fission matrices for use in a computational fluid dynamics (CFD) code to model transients [62].

2.3 Delayed Neutron Precursors

2.3.1 Overview

An important consideration in nuclear reactors, especially in MSRs, are DNPs, discovered in 1939, with initial spectral measurements in 1946 [81, 28, 22]. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [9]. In this work, the term DNP will be solely used, as the delay of the neutron emission practically is only due to the beta decay [? ?]. Classifying between the emitter and the DNP is not relevant for this work.

This delay in the emission of a neutron has a large effect on the reactor period and transient behavior. The time-dependent rate of delayed neutron emission from a single beta-delayed neutron emitting nuclide, or DNP, is shown in Equation (2.21).

$$n_{d,i}(t) = P_{n,i} \lambda_i N_i(t) \quad (2.21)$$

The terms in Equation (2.21) is defined as follows:

- $n_{d,i}$ is the delayed neutron activity, or count rate, from DNP i
- $P_{n,i}$ is the average number of delayed neutrons emitted from a single beta decay of nuclide i , or the emission probability

- λ_i is the decay constant of DNP i
- $N_i(t)$ is the concentration of DNP i

A challenging aspect of treating DNPs in MSRs is the necessary inclusion of a spatial component due to the movement of the fuel salt. In a solid fuel reactor, such as an LWR, a spatial component is not necessary to properly model the DNPs. Modeling space and time together causes computational costs to increase significantly, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels. Although it is challenging, it is important to properly model the DNPs in reactors.

The DNPs are important to model properly for several reasons. Material accountancy, reactivity worth, decay heat, and reactor controllability (particularly for MSRs) are all key aspects to understand for safe reactor operation [28, 29, 31, 59]. Without properly modeling DNPs, these aspects will not be accurately predicted. This leads to over-conservatism in safety systems and increased costs, difficulties in licensing, and increased risk.

2.3.2 Measurement

Macro Measurement of Delayed Neutron Precursors

There are two approaches used for measurement of DNPs, referred to here as macro and micro measurement approaches. This section covers the macro approach. Analysis and collection of macro approach data is very common in the literature, as the data from this method is important for reactors [54, 97, 68, 12, 100, 19, 49, 66, 67]. The macro approach entails irradiation of a sample containing a single fissile nuclide at or near a given neutron energy. Small samples are used in order to minimize neutron multiplication within the sample. Two different types of irradiation are used: pulse irradiation, where the irradiation time is shorter than the shortest half-life; and infinite, where the irradiation time is longer than the longest half-life. After irradiation, the delayed neutrons from the sample are counted as a function of time. Equation (2.22) shows the delayed neutron counts as a function of time after irradiation.

$$n_d(t) = \epsilon \sum_{i=1}^I P_{n_i} \lambda_i N_i(t) \quad (2.22)$$

Many of the terms in Equation (2.22) are the same as in (2.21). The terms in Equation (2.22) which are different are defined as follows:

I is the total number of DNPs

ϵ is the detector efficiency

Once this delayed neutron temporal profile is generated, it is common to then use the data to generate group and kinetics parameters for that fissile nuclide and neutron energy. The specifics of the methods used to form those parameters is discussed in Section 2.3.3. Figure 2.1 (recreated from [54]) shows how an experimental setup for collecting delayed neutron counts works.

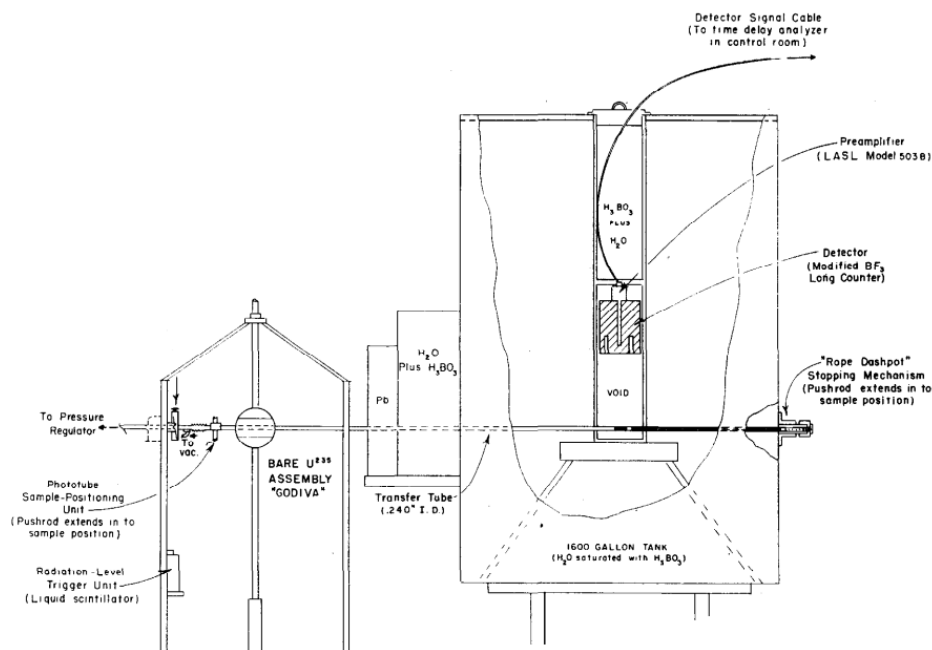


Figure 2.1: Experimental setup for measuring delayed neutron emission from irradiation of GODIVA on a sample recreated from [54].

There are challenges associated with the collection of this experimental data [9]. Experimental noise becomes more significant after long periods of time, as the delayed neutron counts decrease. Additionally, the short lived groups cannot be measured directly due to moderated neutrons from irradiation. This means that both the long and short time-scale measurements have greater uncertainty than the middle time-scale measurements.

Micro Measurement of Delayed Neutron Precursors

The other approach used to measure DNPs is the micro-measurement approach. This approach entails taking a sample of a single DNP nuclide and gathering experimental data on its properties. Typically this involves some form of rapid chemical extraction or isotope separator and magnetic tape in order to rapidly measure short lived nuclides.

There are many different experimental methods used to study DNPs, particularly their spectra: on-line mass spectrometry, helium-3 spectrometry, the proton-recoil technique, the time-of-flight method, and the neutron scintillator [28]. On-line mass spectrometry combined with a helium-3 ionization chamber and the proton-recoil

technique allows for measurement of the delayed neutron energy spectrum [91, 16]. The time-of-flight method involves measuring the velocity of the delayed neutron by having a known travel distance and time measurement [26, 41, 25]. An example of the time-of-flight experimental setup is shown in Figure 2.2.

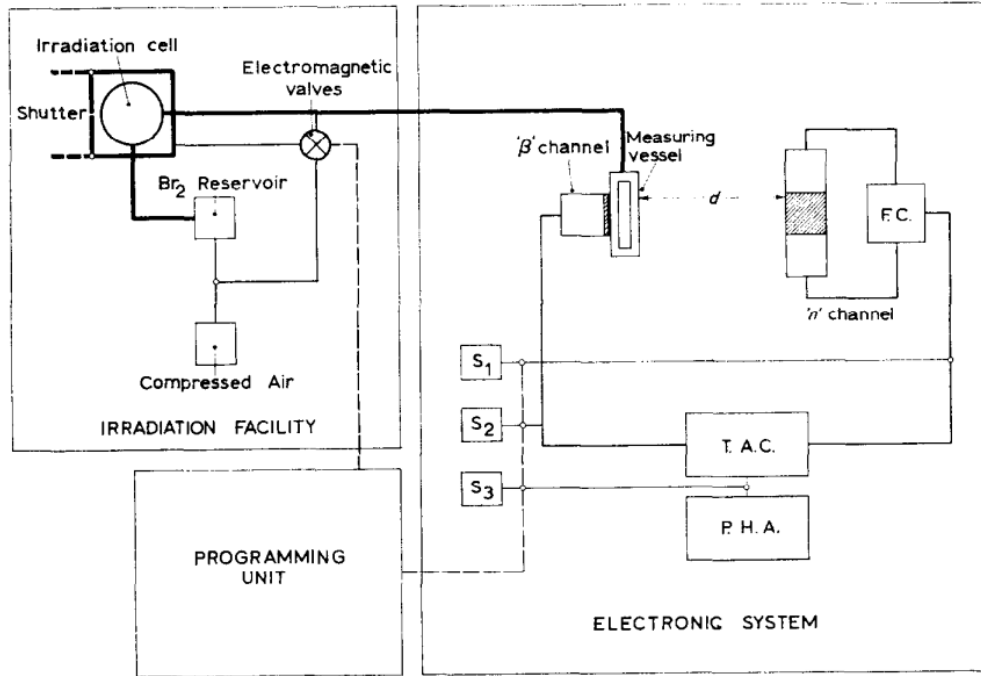


Figure 2.2: A block diagram of the time-of-flight experimental setup for measuring delayed neutrons from bromine-87 and bromine-88, extracted from the “Democritos” reactor via a fast radiochemical technique recreated from [25].

This micro measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis, rather than macro-scale. This is particularly important for MSRs, where the spatial concentration of DNPs depends on half-lives, yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades on these nuclides. Fortunately, the IAEA has worked to compile and evaluate this data into a publicly accessible form [33, 32]. This data, when combined with cumulative fission yields and neutronics data, can be used to model the DNPs explicitly, rather than using groups.

2.3.3 Group and Kinetics Parameters

Overview

In Section 2.3.2, the various methods for experimentally understanding delayed neutron precursors are discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well

as the various ways they are represented. The common way of representing the DNP groups is in several distinct groups, typically six. These groups were developed originally for computational efficiency and due to lacking data for individual precursors. Brady and England found that increasing the number of groups for point kinetics calculations caused no significant differences [18]. This shows that merely increasing the number of DNP groups does not provide an increase in accuracy for point kinetics simulations.

The DNP groups come with associated parameters, such as the group decay constants, λ_i , and the group abundances, a_i . These group abundances can be converted to group yields, β_i , by dividing the group abundances by the average number of neutrons emitted per fission event. The group abundances represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_i and λ_i terms are the defining parameters for the DNP groups, along with the group spectra, χ_g . With this data, the other group parameters can be derived, such as the total delayed neutron yield, Y , and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNPs nuclides are needed. For example, in 1986, Perry et al. demonstrates that kinetics calculations yield different results when using individual nuclides compared to the six group fit data [77]. In 1989, Brady and England also use the individual nuclides to compare with group data [18], and show that the updated six group parameters offer consistent results with the explicit DNPs. By updating the six group parameters based on the updated data available, greater consistency between experimental results and simulations was made possible.

With the DNPs represented in group form, it also simplifies the calculation of various kinetics parameters. Various kinetics parameters are introduced in Section 2.2.1, though without detail on the purpose of each variable. This section provides more information on what these kinetics and delayed neutron group parameters physically represent, while Sections 2.3.3 and 2.3.3 detail how they are calculated.

Arguably the most well known term in nuclear engineering is the effective multiplication factor, k_{eff} . This term represents the scaling of the number of neutrons over each generation. There is also the infinite multiplication factor, k_{∞} , which represents the multiplication without considering leakage terms. The trend of the effective form of the term requiring information on the specific reactor geometry continues for the delayed neutron fraction term.

The delayed neutron fraction, β , represents the average number of delayed neutrons emitted per fission event. Alternatively, it can be thought of as representing the ratio of delayed neutron production over all neutron production [72]. This term is directly related to the DNPs, and can be constructed from the DNP groups.

Related to the delayed neutron fraction is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it is dependent on the properties of the reactor core. This term represents the ratio between the adjoint neutron flux and spectrum-weighted delayed-

neutron induced fission to the total number of fission events [86]. Alternatively, it can be thought of as the ratio of the spectrum and adjoint weighted delayed neutron production over all spectrum and adjoint weighted neutron production [72]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

Another term that is important for reactor kinetics is the mean neutron generation time, Λ . The mean neutron generation time is the average "...time between the birth of a neutron and the subsequent absorption that induces fission" [86]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [98]. In contrast, the effective delayed neutron fraction varies by on the order of 20% from the delayed neutron fraction [98].

The adjoint flux is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather serves as an importance function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection.

Calculation of group parameters

The calculation of the kinetics parameters depend on the DNP group parameters, which themselves are typically formulated based on experimental data as previously discussed.

Equation (2.23) shows the exact formulation for the delayed neutron counts as a function of time measured by a detector for I individual DNPs.

$$n_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t) \quad (2.23)$$

Equation (2.24) instead shows the n group DNP fit measured by the same detector for a pulse irradiation. For an infinite, or saturation, irradiation, Equation (2.25) is used [19]. To make this equation generic, the $a_i \lambda_i$ term and a_i term can instead be written as A_i . Additionally, the fission term can be written as $\bar{F}_s$, to represent either a fission rate or a net number of fissions.

$$n_d(t) \approx \epsilon \bar{F}_s \sum_{i=1}^n a_i \lambda_i e^{-\lambda_i t} \quad (2.24)$$

$$n_d(t) \approx \epsilon \dot{F}_s \sum_{i=1}^n a_i e^{-\lambda_i t} \quad (2.25)$$

The terms in Equation (2.23), (2.24), and (2.25) are defined as follows:

I is the total number of DNPs

n is the number of DNP groups

ϵ is the detector efficiency

$P_{n,i}$ is the emission probability of DNP i

λ_i is the decay constant of the i th DNP or group

$N_i(t)$ is the time dependent number of atoms of nuclide i

F_s is the total number of fission events in the pulse irradiation

$\dot{F}_s$ is the fission rate for a saturation irradiation

a_i is the i th group yield, or group abundance

Once the delayed neutron counts are acquired, an over-determined non-linear least squares approach is used to calculate the group abundances and decays. For example, Brady used STEPIT to handle the non-linear least-squares solve with both pulse and infinite irradiation experimental data [19]. Waldo et al. also solve the least-squares problem, though the specific approach used is not specified [100]. The general form of the solve is given in Equation (2.26).

$$A\vec{x} \approx \vec{b} \quad (2.26)$$

Within this least squares equation, the solution vector contains only the group abundances, shown in Equation (2.27).

$$\vec{x} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \quad (2.27)$$

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.24) and (2.25). Equation (2.28) shows the coefficient matrix for a pulse irradiation, though for a saturation irradiation the only change would be removing the λ_i term.

$$A = \begin{pmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{pmatrix} \quad (2.28)$$

The target vector contains the delayed neutron data normalized to the number of fissions, or fission rate, and the detector efficiency, as shown in Equation (2.29).

$$\vec{b} = \begin{pmatrix} n_d(t_0)(\epsilon F_s)^{-1} \\ n_d(t_1)(\epsilon F_s)^{-1} \\ \vdots \\ n_d(t_m)(\epsilon F_s)^{-1} \end{pmatrix} \quad (2.29)$$

By solving this non-linear system of equations, the group half lives and abundances can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, k , then Equation (2.30) can be used to determine the total group half lives [28].

$$\lambda_i = \frac{\sum_k \lambda_i^{(k)} \beta_{i,eff}^{(k)}}{\sum_k \beta_{i,eff}^{(k)}} \quad (2.30)$$

These equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Equation (2.31) shows that including the energy in the delayed neutron count requires the energy spectra, $\chi_i(E)$, of the relevant group to be included. This is also the general form of the saturation and pulse irradiation equations, as it includes A_i , which can represent $a_i \lambda_i$ or a_i depending on the irradiation type. The $\tilde{F}_s$ value also represents that this could be the total number of fissions, for a pulse irradiation, or fission rate, for saturation irradiation.

$$n_d(E, t) \approx \epsilon \tilde{F}_s \sum_{i=1}^n \chi_i(E) A_i e^{-\lambda_i t} \quad (2.31)$$

A similar procedure as discussed for no energy spectra modeling can be used in order to calculate the DNP group energy spectra.

Another consideration in calculating group parameters is constructing group concentrations from individual

nuclide data. This is less important for group parameters, but is important if there is desire to use the generated group parameters with concentrations which are converted into group concentrations.

Two different methods are considered for the separation and recombination of DNP groups into individual nuclides. These methods have been used for generation of DNP spectral fitting:

- Grouping based on half lives [35, 87]
- Grouping based on fractional contribution to adjacent groups [19]

The grouping based on half lives is easier to implement and understand, as it simply requires comparing each nuclide's half life to a constant value. The groupings based on half lives are given in the top two sets in Table 2.2. The bottom set of half lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach.

Table 2.2: Half-lives of DNP groups.

Source	Grouping [s]
[35]	$T_1 > 31$
[35]	$31 \geq T_2 > 9$
[35]	$9 \geq T_3 > 3$
[35]	$3 \geq T_4 > 0.8$
[35]	$0.8 \geq T_5 > 0.3$
[35]	$0.3 \geq T_6$
[87] [*]	$T_1 > 30$
[87]	$30 \geq T_2 > 10$
[87]	$10 \geq T_3 > 4$
[87]	$4 \geq T_4 > 1.4$
[87]	$1.4 \geq T_5 > 0.4$
[87]	$0.4 \geq T_6$
[21]	$T_1 = 51.98$
[21]	$T_2 = 21.17$
[21]	$T_3 = 5.74$
[21]	$T_4 = 2.29$
[21]	$T_5 = 0.82$
[21]	$T_6 = 0.24$

^{*} Rudstam uses $T_1 = 55.6$, but for the sake of completeness, this has been modified

The half-life grouping method thus results in a series of comparisons. If a nuclide is within a given group's bounds, the concentration of that nuclide is then added to that group.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups for each nuclide k , which are groups i and $i + 1$. These groups are determined via Equation (2.32). Nuclides which do not have a λ_{i+1} which is larger than λ_k will have the entire fractional contribution to group i .

$$\lambda_i < \lambda_k < \lambda_{i+1} \quad (2.32)$$

The fractional contribution of nuclide k to each group is represented with $f_{k,i}$ and $f_{k,i+1}$. The total fraction must be equal to 1 as shown in Equation (2.33).

$$f_{k,i} + f_{k,i+1} = 1 \quad (2.33)$$

To solve for the fractions of each group each nuclide contributes to, the least squares minimization problem is solved. This is shown in Equation (2.34).

$$\int_0^\infty \left(\lambda_k e^{-\lambda_k t} - f_{k,i} \lambda_i e^{-\lambda_i t} - f_{k,i+1} \lambda_{i+1} e^{-\lambda_{i+1} t} \right)^2 dt \quad (2.34)$$

Solving this equation for where the partial derivatives are zero for each group fraction, combined with Equation (2.33), yields the overdetermined system of equations shown in (2.35).

$$A = \begin{bmatrix} \lambda_i & \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} \\ \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} & \lambda_{i+1} \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} \frac{2\lambda_i \lambda_k}{\lambda_i + \lambda_k} \\ \frac{2\lambda_k \lambda_{i+1}}{\lambda_k + \lambda_{i+1}} \\ 1 \end{bmatrix}, \quad x = \begin{bmatrix} f_{k,i} \\ f_{k,i+1} \end{bmatrix} \quad (2.35)$$

This system of equations is then solved using the pseudo-inverse, shown in Equation (2.36).

$$x = (A^T A)^{-1} A^T b \quad (2.36)$$

With the fractional contribution vector, the concentration of a given nuclide can then be multiplied by each fraction to show much it contributes to that group. With a running summation, the group concentrations can thus be calculated.

Calculation of kinetics parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. Equation (2.37) shows that to calculate the group yield, the group abundances are divided by the average number of neutrons emitted from each fission event.

$$\beta_i = \frac{a_i}{\bar{\nu}} \quad (2.37)$$

Equation (2.38) shows how the average number of delayed neutrons per fission event, $\bar{\nu}_d$, can be calculated by summing the group abundances [42]. Based on the definition of β_i given in Equation (2.37), it can be seen that the group abundances, a_i , could be rewritten as the group average delayed neutrons per fission event, $\bar{\nu}_{d,i}$.

$$\bar{\nu}_d = \sum_i a_i \quad (2.38)$$

The delayed neutron fraction can be written using the ratio of the average delayed neutrons per fission over the average neutrons per fission. This is shown in Equation (2.39).

$$\beta = \frac{\bar{\nu}_d}{\bar{\nu}} \quad (2.39)$$

The mean neutron generation time can be calculated using Equation (2.40), where l is the prompt neutron lifetime and k is the multiplication factor. This can be calculated using a time-dependent version of a Monte Carlo code along with PRKs [86].

$$\Lambda = \frac{l}{k} \quad (2.40)$$

Finally, there are the effective parameters: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . β_{eff} can be calculated explicitly by its definition, which is "...the ratio between the adjoint neutron flux and spectrum-weighted delayed-neutron induced fission to the total number of fission events." This is shown in Equation (2.41) [86].

$$\beta_{eff} = \frac{\int \phi^\dagger \chi_d \beta \nu \Sigma_f \phi \, dE d\vec{\Omega} d\vec{r}}{\int \phi^\dagger \chi \nu \Sigma_f \phi \, dE d\vec{\Omega} d\vec{r}} \quad (2.41)$$

The terms in Equation (2.41) are defined as follows:

$\phi^\dagger$ is the adjoint neutron flux

χ_d is the delayed neutron energy spectrum

χ is the fission neutron energy spectrum

β is the delayed neutron fraction

ν is the average number of neutrons released per fission

Σ_f is the macroscopic fission cross section

ϕ is the neutron flux

Λ_{eff} can also be calculated explicitly by its definition, which is the "...time between the birth of a neutron and the subsequent absorption that induces fission." This is shown in Equation (2.42) [98].

$$\Lambda_{eff} = \frac{\int \phi_0^\dagger \frac{1}{\nu} \phi_0 dE d\vec{\Omega} d\vec{r}}{\int \phi_0^\dagger F_0 \phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.42)$$

The terms in Equation (2.42) are defined as follows:

$\phi_0^\dagger$ is the critical adjoint neutron flux

ϕ_0 is the critical neutron flux

ν is the neutron speed

F_0 is the total fission operator

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [98]. These include the direct integration, or adjoint weighted, method [55]; the eigenvalue, or perturbation, method [46]; and the iterated fission probability method [17]. There is also the pulsed method for calculating Λ_{eff} , and the prompt method for calculating β_{eff} [86]. For β_{eff} , there are also the Meulekamp and Spriggs methods which can be used [72]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods, which show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the MSFR [7].

2.4 Molten Salt Reactor Depletion

2.4.1 Overview

There are many different MSR designs in the literature, including [24]: the FUJI series [40], Actinides Molten Salt TransmutER (AMSTER) [99], MOSART [51], Fast Spectrum Molten Salt Reactor (FS-MSR) [47], European MSFR [70], Small Mobile Molten Salt Reactor (SM-MSR) [23], MSBR [83], and MSRE [82]. These designs feature a circulating salt

containing fissionable material, which has different characteristics than stationary fuels traditionally used in LWRs. In particular, the circulating fuel salt allows for online chemical reprocessing, causes depletion to vary spatially far more than stationary fuel, movement of DNPs, and close coupling of thermal-hydraulics to neutronics.

There are many works in the literature which have investigated online chemical reprocessing of MSRs, as can be seen in Table 2.3 recreated from [89]. Table 2.3 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, which shows most models do not include DNP movement in depletion. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models and is also sorted by the publication year in the "Year" column. The table shows that there are many MSRs, shown in the "Reactor" column, which have had depletion modeled using a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.3: Molten Salt Reactor Models Analyzed with Methods Approximating Online Reprocessing [89]

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [53]	Continuous	Unit Cell	N/A	No	2020
Zhuang et al. [107]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [102]	Continuous	Full 3D	N/A	No	2019
Zhou et al. [106]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [8]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [84]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [88]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [15]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [79]	Batchwise	Full 3D	Toy	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [76]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [92]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [75]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [37]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [73]	Mixed	Unit Cell	Toy	Yes	2008
Merle-Lucotte et al. [69]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [44]	Mixed	2D Slice	N/A	Yes	1971

Although these works incorporate the effects of online chemical reprocessing on depletion, an effect they do not explicitly model is the spatial effect on depletion. The reason why most works do not include spatial resolution in depletion calculations is due to the computational cost.

2.4.2 Depletion Simulations

Depletion itself is the process of fission, transmutation, decay, reprocessing, and the changes to material compositions this process causes. Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [63]. Simulating depletion entails numerically solving the Bateman equations; a generalized form of which is in Equation (2.43).

$$\frac{dN_j}{dt}_{base} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \Phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \Phi) N_i] - (\lambda_j + \sigma_j \Phi) N_j \quad (2.43)$$

The terms in Equation (2.43) are defined as follows [63]:

N_j is the atomic density of isotope j $\left[\frac{atoms}{cm^3} \right]$

$\gamma_{i \rightarrow j}$ is the fractional fission product yield of j in the fission of isotope i

$\sigma_{f,i}$ is the microscopic fission cross-section of isotope i $[cm^2]$

Φ is the spectrum-averaged scalar flux in the fuel region $\left[\frac{neutrons}{cm^2 s} \right]$

$\lambda_{i \rightarrow j}$ is the decay constant of decay $i \rightarrow j$ $[s^{-1}]$

$\sigma_{i \rightarrow j}$ is the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[cm^2]$

N_i is the atomic density of isotope i $\left[\frac{atoms}{cm^3} \right]$

λ_j is the decay constant of isotope j $[s^{-1}]$

$\lambda_{r,j}$ is the reprocessing constant for removal of isotope j $[s^{-1}]$

σ_j is the microscopic total transmutation cross-section of isotope j $[cm^2]$

$\lambda_{f,j}$ is the feed constant for feed of isotope i $[s^{-1}]$

The matrix form can be seen in (2.44), where N is a vector of compositions, and A is the depletion matrix. This matrix contains the gain and loss terms for each nuclide in N , which vary continuously as a function of time and depend on the concentrations as well. Further documentation on treatment of depletion of these terms can be found in the OpenMC documentation, an open source Monte Carlo transport code which has built-in depletion solvers [85].

$$\frac{dN}{dt} = AN \quad (2.44)$$

There are several approaches to solving this equation, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [64]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [75] to implement Fourth order Runge–Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [8].

Overall, there are many approaches to solving the depletion equation with different computational costs and levels of accuracy. Building a code to handle depletion requires two components: the depletion matrix and a numerical solver.

Stationary fuel

For a stationary fuel, the Bateman equation as shown previously is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten to the form shown in Equation (2.45). Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. This also means that the loss term, $\lambda_{net,j}$, does not vary with time in this equation.

$$\frac{dN_j}{dt} = C(t) - \lambda_{net,j}N_j(t) \quad (2.45)$$

The terms in Equation (2.45) are defined as follows:

C is the source term for nuclide j , containing fission, decay, and transmutation sources

$\lambda_{net,j}$ is the loss term for nuclide j , containing decay and transmutation losses

When solving this equation, the common approach is to have different material blocks in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, no spatial component needs to be considered, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel

Circulating fuel makes the problem more complex, as adding a spatial component means that instead of a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static

Bateman equation can be modified to include advection and diffusion terms, yielding Equation (2.46). In this equation, the assumption is made that only concentration spatial resolution is applied, so the depletion matrix does not include additional spatial resolution. Thus, only the concentrations and material blocks provide spatial resolution to the problem. More spatial refinement can be provided to the neutronic parameters based on the specific approach used, such as a fully spatially resolved neutron flux.

$$\frac{dN_j}{dt} = C(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)) \quad (2.46)$$

The terms in Equation (2.45) contain several repeated terms from Equation (2.45), with new terms defined as follows:

D_{mix} is the mixing diffusivity, which sums mass diffusivity and turbulent diffusion

$\vec{u}$ is the fluid velocity field

As previously mentioned, the main challenge faces when solving this problem is the computational cost. However, there are still many works which have attempted to develop methods to get around this problem.

One approach is very computationally inexpensive, and is referred to as the "scaled flux" method [14]. However, this approach does not model the in-core and ex-core regions in a physical manner. Instead, the scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. Another approach is to calculate the steady-state concentration, changing the system of ordinary differential equations to spatial rather than temporal relations [90].

Other approaches, such as Monte Carlo N-Particle (MCNP) and Advanced Dimensional Depletion for Engineering of Reactors (ADDER), or the method proposed by Tano et al. using Griffin and Pronghorn, try to model the flow more explicitly [6, 96]. This approach is more computationally expensive than the other approaches that only require solving a system of ordinary differential equations. However, these works have shown that including advection in depletion directly affects the equilibrium concentration of xenon-135 by roughly 70% [96, 36]. This means that depletion simulations of MSRs need to include spatially resolved depletion to generate accurate results.

However, including spatial resolution in depletion has been referred to as computationally "intractable" [14, 96]. This is because tracking the ~2000 nuclides over space and time requires new methods rather than those used for the time dependent system of ordinary differential equations. Methods for solving the system of partial differential equations require accurately capturing short time scale flow effects, which makes multi-year long depletion simulations exceedingly expensive. Numerical stability of the problem, when using an explicit-time solving scheme, also requires that the Courant-Friedrichs-Lewy (CFL) condition shown in Equation (2.47) to be valid.

$$\Delta t \leq \lambda_{CFL} \frac{\Delta z}{v} \quad (2.47)$$

This equation limits the time step size based on the flow rate and spatial mesh size. Although using a different time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, resolution of spatial mesh, and the overall amount of time simulated in depletion.

Chapter 3

Tools

3.1 Data

JANIS and IAEA

3.2 OpenMC

3.2.1 Data

3.3 MOOSE Apps

3.3.1 Griffin

3.3.2 ISOXML

3.3.3 Pronghorn

Chapter 4

Methodology

4.1 Flow Tabulation Simplification

Discuss how the flow can be represented as residence times and mix rates, generically.

4.2 Resolved Delayed Neutron Counts

Sample needs to be as small as possible to minimize self-multiplication [54].

4.2.1 Delayed Neutron Precursor Concentrations

4.2.2 Emission Probability and Decay Data

4.3 Delayed Neutron Precursor Group Reconstruction

4.3.1 Group Parameters from Counts

Split up into pulse and saturation irradiation (going to need to find more information on how to do this) (two groups of 4 then renormalization?)

Non-linear least-squares problem (this is for getting group params)

4.3.2 Group Concentrations from Nuclides

Fractional fitting (this is for running next Griffin/Pronghorn step)

4.4 Simulated Scenarios

4.4.1 Simplified Model Verification

4.4.2 Safeguards

4.4.3 Safety

Chapter 5

Preliminary Results

5.1 Group Parameter Comparisons

5.2 Spatially Resolved Depletion Effects on Precursors

Net delayed neutron yield; group abundances; group decays; comparison of most impactful nuclides (?)

5.3 Verification

5.3.1 Simplified Model

5.3.2 No Molten Properties Comparisons

No flow, reprocessing, thermochemistry, spatial depletion should equal expected results.

5.3.3 Literature Comparison

Should match other Griffin/Pronghorn results

5.4 INL work

This work uses Griffin/Pronghorn/Thermochemica to simulate safeguards diversion while tracking all DNPs (but doesn't update DNP groups, only concentrations).

Chapter 6

Proposed Work

6.1 Resolve DNP spectra

[28] discusses why it's important in section 6.

6.2 Uncertainty Quantification

6.3 Finish this work

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